Practical Test information

This practical test relates to the following Unit Learning Outcome (ULO):

- ULO 2: Apply computer hardware and office information system to prepare reasonably complex professional documents.

In this challenge, you will apply your skills in a single workbook. Each task is organised on a separate worksheet, and it is crucial to follow the specific instructions for each task carefully.

1. You will be working on a single Excel workbook for this challenge. This means all your tasks will be done in one file.

2. This challenge consists of **three tasks**. Specific instructions are provided for each task as you navigate through them. Please read them carefully.

3. You are to create your own working file in ***.xlxs*** file format for this challenge is provided.

4. Format the data as shown in the figure for each task. This may involve changing font size, styles, cell colours, or alignment.

5. Periodically save your work to prevent data loss.

6. Rename the working file to **Student ID_Student Name_MSExcelChallenge.xlsx**. For example, *7432593_Gloria Law_MSExcelChallenge.xlsx*.

7. Once completed, show the document to your tutor for verification and grading.

Challenge Tasks

Challenge Task 1: 2

Challenge Task 2: 4

Challenge Task 3: 5

Challenge Task 1:

1. Open a new excel file, and rename the worksheet to ***Service Revenue***. Format the worksheet as below:

	A	B	C	D	E	F	G	H	I	J
1	***Revenue Report: Second Half of the Year***									
2	**Services**	**July**	**August**	**September**	**October**	**November**	**December**	**Total**	**Revenue Average**	**Rank by Average**
3	**Catering**	RM2,335.00	RM2,500.00	RM3,356.00	RM4,128.00	RM4,223.00	RM7,453.00			
4	**Logistics**	RM8,130.00	RM5,932.00	RM7,462.00	RM6,123.00	RM9,700.00	RM3,547.00			
5	**e-Hailing**	RM5,241.00	RM4,320.00	RM6,335.00	RM5,241.00	RM4,418.00	RM3,232.00			
6	**Delivery**	RM1,569.00	RM1,925.00	RM2,241.00	RM2,670.00	RM3,034.00	RM3,516.00			
7	**Printing**	RM4,160.00	RM3,350.00	RM6,571.00	RM6,001.00	RM5,794.00	RM5,462.00			
8	**Recruitment**	RM6,521.00	RM6,000.00	RM5,139.00	RM4,322.00	RM4,100.00	RM3,357.00			
9	**Monthly Total**									
10	**Monthly Average**									
11										
12			**Amount**	**Service Type**						
13	**Highest Average Revenue**									
14	**Lowest Average Revenue**									
15										
16	**Number of services with average revenue less than** RM 5,000 **:**									

2. Merge the cells A1:J1.

3. Format the fonts to the following cells:
   - A1:J1 – Calibri, 16 pt, bold and italics
   - A2:J2 – Calibri, 11 pt, bold
   - C12:D12 – Calibri, 11 pt, bold
   - The rest of the worksheet – Calibri, 11pt.

4. Bold column A.

5. Use an appropriate function or formula to find the total revenue for each service in cells H3:H8, and monthly total in cells B9:G9.

6. Use an appropriate function to find the revenue average for each service in cells I3:I8, and monthly average in cells B10:G10.

7. In cell J3, use an appropriate function to rank the services based on their revenue average. The service with the lowest revenue average is ranked #1. Apply absolute reference to the formula.

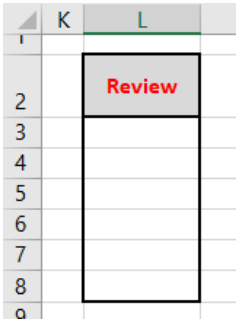
8. Sort the services based on their Rank by Average in ascending order.

9. Use an appropriate function to find the **highest and lowest average revenue** amongst the services in cells C13 and C14 respectively.

10. In cell D13 and D14, insert the formula using the `INDEX-MATCH` function to display the type of service that corresponds to the highest average revenue in cell C13 and C14 respectively. Apply absolute reference to the formula.

11. Find the number of services with an average revenue of less than RM 5,000 in cell E16.

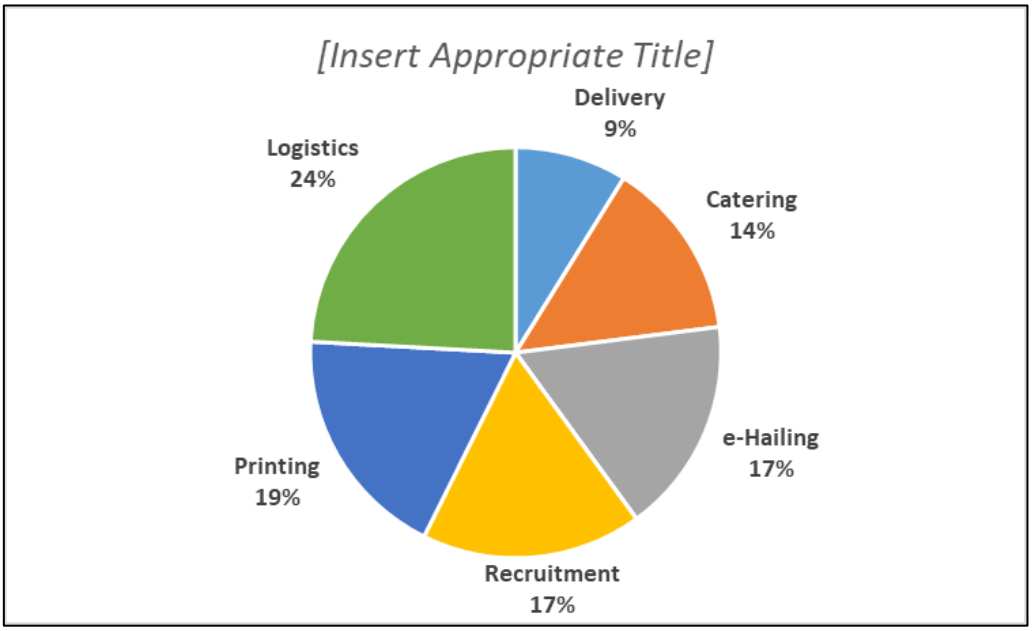
12. From cell L2 to L8, create the following box. **Centre-align cells L2:L8.**



13. In cell L3, insert the formula using an appropriate `IF` function to which services need review. Services should be labelled "Review" if the average (cell I3) is under RM5,000. Otherwise, label it as "Keep". Repeat for the remaining cells.

14. Format **all** the currency values with a RM symbol and 2 decimal places.

15. Create a 2-D Pie Chart with an appropriate title to show the average revenue for all services. Remove the legends and format the data label to display "Category Name", "Percentage" and "Show Leader Lines". Set the Label Position to "Outside End". Your pie chart should be similar to the one below.



16. Place the charts in the same worksheet below the data table.

17. Save your work periodically to prevent data loss.

Challenge Task 2:

1. In the same excel document, create a new worksheet with the name ***Employee Attendance***. Format the worksheet as below:

	A	B	C	D	E	F	G
1	**Company A Employee Attendance Record (22nd September 2026)**						
2	*Date and Time:*	*9/22/2026 19:04*					
3							
4	**Employee ID**	**Employees Name**	**Department**	**Scheduled Start Time**	**Clock-In Time**	**Punctuality Status**	**Late Deduction (RM)**
5	SS1234	Claire Hamlin	Research	10:00 AM	10:05 AM		
6	SS0467	Hermon Brown	Sales	8:30 AM	8:15 AM		
7	SS0024	Michael Smith	Operations	8:30 AM	8:29 AM		
8	SS1023	Nicole Gates	Operations	12:30 PM	12:31 PM		
9	SS0068	James Morrison	Sales	8:30 AM	9:43 AM		
10	SS1199	Ashley Price	Research	10:00 AM	9:37 AM		
11	SS0813	David Bullock	Operations	7:30 PM	7:49 PM		
12	SS1001	Jane Ashlee	Research	12:00 PM	11:00 AM		
13							
14		**Data Summary**			**Employee Attendance Search**		
15		**Late Employees**			**Employee ID:**		
16		**On-Time Employees**			**Employee Name:**		
17					**Department**		
18					**Punctuality Status:**		

2. Merge the cells A1:G1. Bold the words in these cells.

3. Insert the formula for the current date and time in cell B2. Apply “Thick Outside Borders” to the same cell.

4. Format the fonts to the following cells:
   - A1:G1 – Calibri, 16 pt
   - The rest of the worksheet – Calibri, 11pt.

5. In cells F5:F12, use the `IF` function to determine if the employee is late or on time.
   - If the Clock-in Time is after the Scheduled Start Time, return “LATE”.
   - Otherwise, return “ON TIME”.

6. In cells G5:G12, use the `IF` function to decide if a late penalty should be implemented based on cells F5:F12. If the employee is late, apply a RM15 deduction, otherwise RM0.

7. In cell C15, use an appropriate function to count the number of late employees.

8. In cell C16, use an appropriate function to count the number of employees who are on time.

9. The Employee Attendance Search is from cells E14:F18. In cell F15, use the data validation feature to create a dropdown list for employee ID.

10. In cells F16:F18, use the `INDEX-MATCH` function to identify the employee name, department, and punctuality status based on the employee ID in cell F15.

11. Save your work periodically to prevent data loss.

Challenge Task 3:

1. In the same excel document, create a new worksheet with the name ***e-commerce***. Format the worksheet as below:

Product Catalogue

Item Code	Product Name	Category	Unit Weight (kg)	Unit Wholesale (RM)	Retail Price (RM)	Profit Margin (%)	Stock Level	Reorder Point	Reorder Alert
SKU-101	Ergonomic Desk	Furniture	15.5	RM350.00	RM599.00		4	5	
SKU-102	Wireless Mouse	Electronics	0.2	RM25.00	RM49.00		45	10	
SKU-103	4K Monitor	Electronics	6.8	RM450.00	RM799.00		8	10	
SKU-104	Standing Mat	Accessories	2.1	RM40.00	RM89.00		20	50	
SKU-105	Mechanical Keyboard	Electronics	1.1	RM80.00	RM149.00		15	10	
SKU-106	Gaming Headset	Electronics	0.4	RM60.00	RM119.00		23	25	
SKU-107	1080p Webcam	Electronics	0.3	RM50.00	RM99.00		22	22	
SKU-108	32" x 12" Mouse Pad	Accessories	0.4	RM15.00	RM35.00		11	10	
SKU-109	Dual Monitor Arm	Accessories	4.2	RM90.00	RM169.00		32	30	
SKU-110	Ergonomic Office Chair	Furniture	18.0	RM280.00	RM499.00		22	21	

2. Format the fonts to the following cells:
   - A1:G1 – Calibri, 16 pt, Bold.
   - The rest of the worksheet – Calibri, 11pt.

3. Insert the correct formula profit margin in cells G4:G13. Profit margin (%) is calculated using the following formula:

$$Profit\ margin\ (\%) = \frac{Retail\ Price - Unit\ Wholesale}{Retail\ Price} \times 100\%$$

4. In cells J4:J13, use the `IF` function to determine the Reorder Alert status. Display "REORDER" if the stock level is less than or equal to the Reorder Point; otherwise, display "OK".

5. In cells A16:L29, format another table as shown below:

E-Commerce Fulfillment and Inventory Dashboard

System Date: 9/22/2026 19:31

Order ID	Item Code	Product Name	Unit Weight (kg)	Retail Price (RM)	Order Qty	Total Cost (RM)	Total Weight (kg)	Distance (km)	Shipping Tier	Shipping Cost	Total Order Cost (RM)
ORD-501	SKU-101				5			75			
ORD-501	SKU-110				5			75			
ORD-502	SKU-106				20			30			
ORD-502	SKU-104				23			30			
ORD-502	SKU-102				50			30			
ORD-503	SKU-103				10			120			
ORD-504	SKU-107				15			45			
ORD-504	SKU-108				15			45			
ORD-505	SKU-105				5			82			
ORD-505	SKU-107				20			82			

6. In cells B17, use the correct date/time formula to display the **current** date and time.

7. Name the cells A3:J13 ***Prod_catalogue***. Then, in cells C20:E29, use the `VLOOKUP` function to find the product name, unit weight, and retail price based on the product catalogue (***Prod_catalogue***).

8. In cells G20:G29, calculate the total cost of the products. Total cost is the product of retail price and order quantity.

9. In cells H20:H29, calculate the total weight of the products. Total weight is the product of unit weight and order quantity.

10. In cells J20:J29, use the nested `IF` and `AND` functions to determine if the Shipping Tier. The conditions are:

- If the total weight is more than 10kg and the distance is more than 50km, return "Freight".
- If the total weight is less than or equal to 10kg and the distance is less than or equal to 50km, return "Standard"
- Otherwise, return "Express".

11. In cells K20:K29, use the nested `IF` function to calculate the shipping cost. The conditions are:
    - If the shipping tier is "Freight", return 50.
    - If the shipping tier is "Standard", return 30.
    - Otherwise, return 0.
12. In cells L20:L29, calculate the total order cost (RM) of the products. Total order cost is sum of order cost (RM) and shipping cost.
13. In cells A32:C45, format the tables as shown below:

	A	B	C
32	**Total Cost Based on Order**		
33	**Order ID**	**Total Order Cost**	**Cost Ranking**
34	ORD-501		
35	ORD-502		
36	ORD-503		
37	ORD-504		
38	ORD-505		
39	*Grand Total*		

	A	B
41	**Shipping Tier Frequency**	
42	**Shipping Tier**	**Frequency**
43	Freight	
44	Standard	
45	Express	

14. In cells B34:B38, use the `SUMIF` function to calculate the total order cost by adding the costs in L20:L29 when the order ID in cells A34:A38 matches the Order ID in Column A. Apply absolute reference (`$`) to the formula.
15. In cell C34:C38, rank the Total Order Cost in descending order using the `RANK` function.
16. In cell B39, calculate the grand total of total order costs.
17. In cells B43:B45, use the `COUNTIF` function calculate the frequency of each shipping tier listed in A43:A45. You should be referencing cells J20:J29. Apply absolute reference (`$`) to the formula.
18. Generate a column chart based on the data in A33:B38. Your resultant column chart should be similar to the one below.



19. Save your work periodically to prevent data loss.